

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
COMPUTER SCIENCE AND ENGINEERING

VII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22PC1CS401	Big Data Analytics	3	0	0	3	3
22PC1AM402	DevOps	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1CY301	Information and Network Security	3	0	0	3	3
22PE1DS402	Distributed Computing					
22PE1CS401	Advanced Data Bases					
22PE1CS402	Deep Learning					
22PE1AM403	Information Retrieval Systems					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Database Management Systems Stream	3	0	0	3	3
	Algorithm Design Stream					
	Artificial Intelligence and Machine Learning Stream					
	Data Science Stream					
	Ethical AI Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2CS401	Big Data Analytics Laboratory	0	0	2	2	1
22PC2AM402	DevOps Laboratory	0	0	2	2	1
22PW4CS401	Major Project Phase – I	0	0	6	6	3
Total		15	0	10	25	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
COMPUTER SCIENCE AND ENGINEERING

VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1CS403	Computer Vision	3	0	0	3	3
22PE1AM405	Cognitive Engineering					
22PE1CS404	Augmented Reality and Virtual Reality					
22PC1CY401	Blockchain Technology					
22PE1IT403	Adverse Machine Learning					
	PROFESSIONAL ELECTIVE – VI					
22PE1DS405	Time Series Analysis and Forecasting	3	0	0	3	3
22PE1DS406	Quantum Computing					
22PE1CY410	Ethical Hacking					
22PE1AM407	Robotic Process Automation					
22PE1CS405	Social Media Analytics					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4CS402	Major Project Phase – II	0	0	22	22	11
Total		9	0	22	31	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

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ELA – Experiential Learning Assessment
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VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1CS401) BIG DATA ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explore the fundamental concepts of data analytics
- To learn to analyze the data analysis techniques
- To explore the techniques related to mining streams
- To understand, explore Hadoop framework technology and its associated tools along with visualization methods

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify the major sources of data; apply pre-processing tasks and summary statistics on data

CO-2: Understand the concept of Hadoop and analyze data on HDFS and its comparison with other systems

CO-3: Explain Hadoop framework and its components - PIG, HIVE, HBASE, ZOOKEEPER

CO-4: Examine Machine Learning algorithms and represent data using visualization tools

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	1	1	1	1	1	1	1	1	1	3	3	3
CO-2	2	2	2	3	2	2	1	1	1	1	1	1	3	3	3
CO-3	3	3	3	2	3	2	2	2	1	1	1	1	3	3	3
CO-4	3	3	3	3	3	3	2	3	1	1	2	1	3	3	3

UNIT-I:

Data Management: Introduction, Sources of data, Types of Data, Data preprocessing, Data collection and APIs, Exploring and fixing data, Data storage and management, using multiple data sources.

Big Data and Data Analysis: Introduction to Big Data Platform, Challenges of Conventional Systems, Web Data, Evolution of Analytic Scalability, Analytic Processes and Tools, Analysis vs Reporting, Modern Data Analytic Tools.

UNIT-II:

Data Analysis: Introduction, Terminology and concepts, Summary statistics, Central tendencies and distributions, Variance, Distribution properties and arithmetic, Samples/CLT.

Statistical Concepts: Sampling Distributions, Re- Sampling, Statistical Inference - Prediction Error, Regression Modelling, Multivariate Analysis.

UNIT-III:

Mining Data Streams: Introduction to Streams Concepts, Stream Data Model and Architecture, Stream Computing, Sampling Data in a Stream, Filtering Streams.

Hadoop: Meet Hadoop, Comparison with other systems, A brief history of Hadoop and the Hadoop ecosystem, Analyzing the Data with Hadoop,

UNIT-IV:

SQOOP: Hadoop Distributed File System, HDFS concepts, Design of HDFS, Data Flow in HDFS, Developing a Map Reduce Application-How Map Reduce Works

Frameworks: Applications on Big Data Using Pig and Hive – Data processing operators in Pig – Hive services –HiveQL – Querying Data in Hive - fundamentals of HBase and ZooKeeper

UNIT-V:

Data Visualization: Prepare the data for Visualization, Use tools like Tableau, Qlick View and D3, Draw insights out of Visualization tool.

Big Data Ethics and Privacy: Ethical considerations in Big Data Analytics, Privacy concerns and data protection, Case Studies and Applications: Real-world applications of Big Data Analytics. Use cases from various industries (e.g., healthcare, finance, marketing)

TEXT BOOKS:

1. Introduction to Data Mining, Pang-Ning Tan, Vipin Kumar, Michael Steinbach, Pearson
2. Mining of Massive Datasets, Anand Rajaraman and Jeffrey David Ullman, Cambridge University Press, 2012
3. Hadoop: The Definitive Guide, Tom White, 3rd Edition, O'Reilly, 2012

REFERENCES:

1. Making Sense of Data, Glenn J. Myatt, John Wiley & Sons, 2007
2. Big Data Glossary, Pete Warden, O'Reilly, 2011
3. Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, 2nd Edition, Elsevier, 2008
4. Intelligent Data Analysis, Michael Berthold, David J. Hand, Springer, 2007
5. Hadoop: The Definitive Guide, Tom White, 3rd Edition, O'Reilly Media, 2011

ONLINE RESOURCES:

1. <https://www.edureka.co/big-data-and-analytics>
2. <https://nptel.ac.in/courses/106104189>
3. Data Engineering Certification Courses Online - Purdue University Program (simplilearn.com)
4. Learn Big Data with Online Courses, Classes, & Lessons | edX

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1AM402) DEVOPS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce Agile methodologies and DevOps principles, highlighting their role in improving software development efficiency and addressing bottlenecks
- To explore modern software architectures, emphasizing the transition from monolithic to microservices and serverless models while integrating DevOps practices
- To develop understanding in version control and containerization, utilizing Git, GitHub, Docker, and Kubernetes for efficient project collaboration and deployment
- To familiarize with CI/CD and automation skills, covering Jenkins, CircleCI, GitHub Actions, infrastructure as code (Terraform), testing and monitoring tools
- To introduce DevSecOps and security best practices, focusing on secure coding, secret management, container security, and automated security testing

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyse and apply DevOps principles to enhance business agility, and address software development bottlenecks

CO-2: Design and manage software architectures, including monolithic, microservices and serverless using DevOps practices

CO-3: Implement version control and collaboration using Git and GitHub, and manage containerized applications using Docker and Kubernetes

CO-4: Build and manage CI/CD pipelines using Jenkins, CircleCI, and Github Actions for automated deployments and testing

CO-5: Understand IaC and the need to integrate security in DevOps with DevSecOps and GitOps to ensure secure and efficient software delivery

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	2	2	2	3	3	3	3
CO-2	3	3	3	2	3	2	2	1	1	2	2	3	3	3	3
CO-3	3	2	2	2	3	1	1	1	3	2	2	2	3	3	3
CO-4	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	3	3

UNIT-I:

Introduction to DevOps & Business Agility: Introduction to DevOps (What is DevOps, benefits and goals of DevOps, DevOps lifecycle and phases), DevOps and Business Agility (What is business agility, DevOps's role in enhancing agility, collaboration between dev, QA and ops), Bottlenecks (Identifying bottlenecks in software delivery, common bottlenecks, how DevOps addresses bottlenecks).

UNIT-II:

Software Architectures & the Need for DevOps: Introduction to Agile (Agile vs Waterfall model, Agile methodologies – Scrum, Kanban, Continuous Integration and Continuous Development – CI/CD overview, how Agile and DevOps work together), Software Architectures (Importance of separation of concerns, different types of architectures and examples, Monolithic vs Microservices vs Serverless, benefits and challenges of each)

UNIT-III:

Project Management with Version Control, Containerization and Kubernetes: Version Control Systems (Introduction to VCS, Types of VCS, Git vs other VCS tools, Important Git commands, GitHub for collaboration – branching and merging, pull requests, code reviews), Containerization with Docker (What are containers, Docker vs Virtual Machines, building and running containers, port mapping, container networking overview, DockerHub introduction), Kubernetes (What is Kubernetes, core concepts- pods, nodes, deployments, services, Configmaps and secrets, Kubernetes networking overview, scaling applications (horizontal, vertical), persistent storage, deploying applications- kubectl, config files)

UNIT-IV:

CI/CD, Testing and Monitoring: Importance of CI/CD, Jenkins overview (pipelines, build dependencies, plugins, triggers), CI/CD with CircleCI, Using Github Actions for CI/CD, Testing and Automation (importance of automated testing, types of testing – unit, integration, end-to-end, Test Driven Development (TDD), API testing using Postman, JUnits/PyTests), Monitoring and Logging (Importance of logging and monitoring in DevOps, monitoring tools – Prometheus, Grafana, logging tools – Splunk, New Relic, implementing alerts and logging analysis)

UNIT-V:

Infrastructure as Code (IaC), GitOps and DevSecOps Overview: What is IaC and Terraform, writing Terraform configurations, Terraform State Management and Kubernetes infrastructure, GitOps and ArgoCD (brief introduction to GitOps, ArgoCD setup and declarative deployments), DevSecOps – Security in DevOps (Shift-left security & container image scanning, secret management – vault, AWS/Azure secret manager, Automating security in CI/CD)

TEXT BOOKS:

1. Practical Devops, Joakim Verona, 2nd Edition, Ingram short title, 2018
2. DevOps Tools from Practitioner's Viewpoint, Deepak Gaikwad, Viral Thakkar, Wiley

REFERENCE:

1. DevOps: A Software Architect's Perspective, Len Bass, Ingo Weber, Liming Zhu, Addison Wesley

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CY301) INFORMATION AND NETWORK SECURITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand various cryptographic algorithms
- To examine various public-key cryptosystems and the authentication techniques
- To apply the cryptographic primitives to build the secure protocols
- To analyze and design network security protocols
- To Identify and mitigate software security vulnerabilities in existing systems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand basic cryptographic algorithms, message and web authentication and security issues

CO-2: Analyze the various authentication requirements and build secure protocols

CO-3: Analyze various cryptographic primitives and key distribution schemes

CO-4: Examine web security and wireless security

CO-5: Interpret E-mail security and analyze the security of transactions

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	-	-	1	-	1	-	1	-	-	3	2	2
CO-2	3	3	3	2	1	2	-	1	1	3	-	3	3	2	2
CO-3	3	3	1	2	1	1	-	1	1	2	1	2	3	3	2
CO-4	1	3	3	3	3	3	-	3	2	2	2	3	3	3	2
CO-5	1	3	3	2	3	3	-	3	2	2	2	3	3	3	2

UNIT-I:

Introduction: Overview and History of Cryptography, Public and Private-key Cryptography, One-Way Functions, Simple Encryption Schemes: Substitution and Transposition Techniques

Symmetric Key Ciphers: Block Cipher- Definition, Block Cipher Operations, DES, Triple DES, AES. Stream Ciphers- Definition, RC4, RC5.

UNIT-II:

Asymmetric Key Ciphers: Practical Aspects of Public-Key Cryptography, RSA Cryptosystem, Elgamal Cryptosystem, Elliptic Curve Cryptosystems, Merkle-Hellman Knapsack.

Data Integrity: Introduction, Hash Functions-Definition, SHA-512, Message Authentication Code: Authentication requirements, HMAC, CMAC, Digital signatures, Elgamal Digital Signature Scheme.

UNIT-III:

Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public – Key Infrastructure

Advanced Topics: Pseudorandom Generators, Zero-knowledge Proofs.

UNIT-IV:

Transport-level Security: Web security considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH)

Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN, IEEE 802.11i Wireless LAN Security

UNIT-V:

E-Mail Security: Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload, Combining security associations, Internet Key Exchange.

Case Studies on Cryptography and Security: Secure Multiparty Calculation, Virtual Elections, Single Sign On, Secure Inter-branch Payment Transactions.

TEXT BOOKS:

1. Cryptography and Network Security - Principles and Practice, William Stallings, 7th Edition, Pearson Education, 2017
2. Cryptography: Theory and Practice, Douglas Stinson, 3rd Edition or higher, Chapman & Hall/CRC, 2011
3. Cryptography and Network Security, Atul Kahate, 3rd Edition, McGraw-Hill, 2005

REFERENCES:

1. Cryptography and Network Security, C. K. Shyamala, N. Harini, Dr. T. R. Padmanabhan, 1st Edition, Wiley, 2011
2. Cryptography and Network Security, Forouzan Mukhopadhyay, 3rd Edition, McGraw-Hill, 2015
3. Foundations of Cryptography, Oded Goldreich, Cambridge University Press, 2001
4. Information Security, Principles, and Practice, Mark Stamp, Wiley, 2011
5. Principles of Computer Security, W. M. Arthur Conklin, Greg White, Tata McGraw-Hill, 2016

B.Tech. VII Semester

(22PE1DS402) DISTRIBUTED COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce concepts related to distributed computing systems
- To focus on performance and flexibility issues related to systems design decisions
- To expose students to current literature in distributed systems
- To prepare students for a real-world distributed application design

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand models of distributed computing

CO-2: Analyze algorithms for coordination, communication and synchronization in distributed systems

CO-3: Analyze distributed shared memory models

CO-4: Design and implement distributed file systems

CO-5: Design distributed algorithms for handling deadlocks

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	1	-	-	-	-	-	-	-	1	3	3	3
CO-2	2	2	1	1	1	-	-	-	-	-	-	-	3	3	3
CO-3	3	3	2	2	2	-	-	-	-	-	-	-	3	3	3
CO-4	1	1	3	3	2	-	-	-	-	-	1	1	3	3	3
CO-5	-	-	1	1	2	-	-	-	-	-	-	2	3	3	3

UNIT-I:

Distributed Computing Introduction: Definition, Relation to parallel systems, synchronous vs asynchronous execution, design issues and challenges .A Model of Distributed Computations: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Logical Time: Logical clocks, scalar time, vector time, Matrix time, virtual time, Physical clock synchronization – NTP.

UNIT-II:

Global State and Snapshot Recording Algorithms: System model, Snapshot algorithms for FIFO channels, Variations of Chandy-Lamport algorithm, Snapshot algorithms for non-FIFO channels, Snapshots in a causal delivery system, Monitoring global state, Necessary and sufficient conditions for consistent global snapshots, Finding consistent global snapshots in a distributed computation.

Message Ordering and Group Communication: Message ordering paradigms, Group communication, Causal order (CO), Total order, Multicast, Propagation trees for multicast, Application-level multicast algorithms, Fault-tolerant group communication, Multicast algorithms at the network layer.

UNIT-III:

Termination Detection: System model of a distributed computation, Termination detection using distributed snapshots, weight throwing and spanning-tree-based algorithms, Message-optimal termination detection, and Termination detection in a general distributed computing model, Termination detection in the atomic computation model, Termination detection in a faulty distributed system.

Distributed Mutual Exclusion Algorithms: Lamport's algorithm, Ricart–Agrawala algorithm, Singhal's dynamic information-structure algorithm, Lodha and Kshemkalyani's fair mutual exclusion algorithm, Quorum-based mutual exclusion algorithms, Maekawa's algorithm, Agarwal–El Abbadi quorum-based algorithm, Token-based algorithms, Suzuki–Kasami's broadcast algorithm, Raymond's tree-based algorithm.

UNIT-IV:

Deadlock Detection in Distributed Systems: System model, Models of deadlocks, Knapp's classification of distributed deadlock detection algorithms, Mitchell and Merritt's algorithm for the single resource model, Chandy–Misra–Haas algorithm for the AND model, Chandy–Misra–Haas algorithm for the OR model, Kshemkalyani–Singhal algorithm for the P-out-of-Q model .

Distributed Shared Memory: Abstraction and advantages, Memory consistency models, Shared memory mutual exclusion, Wait-freedom, Register hierarchy and wait-free simulations, Wait-free atomic snapshots of shared objects

UNIT-V:

Check Pointing and Rollback Recovery: Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Koo–Toueg coordinated check pointing algorithm, Juang–Venkatesan algorithm for asynchronous check pointing and recovery, Manivannan–Singhal quasi-synchronous check pointing algorithm, Peterson–Kearns algorithm based on vector time, Helary–Mostefaoui–Netzer–Raynal communication-induced protocol.

TEXT BOOKS:

1. Distributed Computing, Ajay D. Kshemakalyani, Mukesh Singhal, Cambridge University Press, 2008
2. Distributed Systems - Principles and Paradigms, Andrew S. Tanenbaum, Maarten Van Steen, Prentice Hall of India, 2004

REFERENCES:

1. Distributed Computing: Fundamentals, Simulations, and Advanced Topics, Hagit Attiya, Jennifer Welch, Wiley
2. Distributed Systems: Principles and Paradigms, Andrew S. Tanenbaum, Maarten van Steen, Prentice Hall of India

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CS401) ADVANCED DATABASES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce distributed database management system and its design issues
- To explore several algorithms for processing queries and be able to use them
- To describe the methods to translate complex conceptual data models into logical and physical database designs
- To demonstrate query optimization and its algorithms
- To enumerate the concepts behind distributed transaction processing

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand data processing and architecture of distributed database

CO-2: Apply partitioning and fragmentation techniques to the data distributed across the network

CO-3: Design and optimize query processing strategies on distributed fragments

CO-4: Explain transaction processing and concurrency control in distributed databases

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	1	-	-	1	2	-	-	-	1	1	2	2	-
CO-2	2	2	3	3	1	1	-	-	1	-	1	1	3	2	-
CO-3	3	3	3	3	3	2	2	-	2	-	2	3	3	3	-
CO-4	2	2	2	2	1	3	2	-	2	-	-	2	3	3	-

UNIT-I:

Introduction: Distributed Data Processing, Distributed Database System, Promises of DDBSs, Problem areas.

UNIT-II:

Distributed DBMS Architecture: Architectural Models for Distributed DBMS, DDMBS Architecture.

Distributed Database Design: Alternative Design Strategies, Distribution Design issues, Fragmentation, Allocation.

UNIT-III:

Query Processing and Decomposition: Query Processing Objectives, Characterization of query processors, layers of query processing, query decomposition, Localization of distributed data.

UNIT-IV:

Distributed Query Optimization: Query optimization, centralized query optimization, Distributed query optimization algorithms.

UNIT-V:

Transaction Management: Definition, properties of transaction, types of transactions. Distributed concurrency control: Serializability, concurrency control Mechanisms & Algorithms, Time stamped & Optimistic concurrency control Algorithms, Deadlock Management.

TEXT BOOKS:

1. Principles of Distributed Database Systems, M. Tamer OZSU and Patuck Valduriez, Pearson Ed., 2001
2. Distributed Databases, Stefano Ceri and Willipse Pelagatti, McGraw-Hill

REFERENCES:

1. Database System Concepts, Henry F. Korth, A. Silberchatz and Sudershan, McGraw-Hill
2. Database Management Systems, Raghuramakrishnan and Johhanes Gehrke, McGraw-Hill
3. Readings in Database Systems, M. Stonebraker, 2nd Edition, Morgan Kauffman, 1993

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CS402) DEEP LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To gain a grasp of the fundamental concepts of deep learning and regularization in the context of deep learning
- To examine approaches for optimization and creating high-performing deep learning networks while taking into account hyper parameters
- To gain understanding of recurrent and recursive networks
- To develop effective solutions for real-world problems utilizing Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Acquire a comprehensive understanding of the fundamental principles underlying Deep Learning

CO-2: Comprehend and implement regularization techniques for neural networks in deep learning

CO-3: Apply optimization techniques to train deep models effectively

CO-4: Acquire a foundational understanding of Convolutional Neural Networks (CNNs), including their basic principles and architecture

CO-5: Investigate various architectures of Recurrent Neural Networks (RNNs) to explore their diverse capabilities and applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	2	1	1	1	2	1	1	1	3	2
CO-2	3	3	1	2	1	2	1	1	1	1	1	1	1	3	2
CO-3	3	3	1	2	1	2	1	1	1	1	1	1	1	3	2
CO-4	3	3	1	2	1	2	1	1	1	1	1	1	1	3	2
CO-5	3	1	2	1	2	1	1	1	1	1	1	1	1	2	2

UNIT-I:

Introduction to Deep Learning: Biological neuron, Importance and Applications of Deep Learning. Issues of Deep Learning, Supervised Learning Networks, Perceptron Networks.

Deep Feedforward Networks: Historical Trends in Deep Learning Deep Feed forward Networks: Example: XOR, Gradient-based learning, Hidden units.

UNIT-II:

Regularization for Deep Learning: Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization under constrained problems, Dataset Augmentation, Noise Robustness, Semi-Supervised Learning, Multi-Task Learning, Early

Stopping, Parameter Tying and Sharing, Sparse Representations, Bagging and Ensemble Methods, Dropout.

UNIT-III:

Optimization for Training Deep Models: How learning differs from Pure Optimization, Challenges in Neural Network Optimization, Basic Algorithms, Parameter Initialization Strategies,

UNIT-IV:

Convolutional Networks: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function.

UNIT-V:

Sequence Modelling: Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks-Teacher forcing and networks with output recurrence, Computing the gradient in a recurrent neural network, recurrent networks as directed graphical models, LSTM, Deep Recurrent Networks, Recursive Neural Networks.

TEXT BOOKS:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville, MIT Press, 2016
2. Neural Networks and Deep Learning, Charu C. Aggarwal, Springer, 2018

REFERENCES:

1. Deep Learning Methods and Applications, Foundations and Trends® in Signal Processing, Li Deng and Dong Yu, Volume 7, Issues 3-4
2. Deep Learning Made Easy with R - A Gentle Introduction for Data Science, Dr. N. D. Lewis, Create Space Independent Publishing Platform, 2016
3. Manning Early Access Program Deep Learning with R Version 1, François Chollet, J. J. Allaire, MEAP Edition, Manning Publications, 2017

ONLINE RESOURCES:

1. <https://www.coursera.org/learn/neural-networks-deep-learning>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1AM403) INFORMATION RETRIEVAL SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To describe the differences between repositories like Database Management Systems, Information retrieval systems and data warehouse
- To discover various pre-processing techniques which can apply on text documents to outline the structure of queries and documents
- To articulate fundamental functions used in information retrieval such as automatic indexing, abstracting, clustering and visualizing data
- To learn the important concepts, algorithms, and data/file structures that are necessary to specify, design, and implement Information Retrieval (IR) systems
- To explore various text search algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify and understand the relationships between various Repository Systems

CO-2: Apply knowledge of data structures and indexing methods in information retrieval Systems

CO-3: Implement various clustering, searching techniques and information visualization techniques

CO-4: Analyze clustering techniques and algorithms using evaluation measures

CO-5: Implement various text search algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	2	3	3	2	1	1	2	2	2	2	2	1
CO-2	3	3	3	3	3	2	2	2	2	2	2	2	3	3	2
CO-3	3	3	3	3	3	3	2	2	2	3	3	2	3	3	3
CO-4	3	3	3	3	3	3	2	2	2	3	3	2	3	3	3
CO-5	2	2	2	2	3	3	2	1	1	2	2	2	3	3	3

UNIT-I:

Introduction: Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses, Information Retrieval System Capabilities – Search, Browse, Miscellaneous.

UNIT-II:

Cataloging and Indexing: Objectives, Indexing Process, Automatic Indexing, Information Extraction

Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N – gram data structure, PAT data structure, Signature file structure, Hypertext data structure.

UNIT-III:

Automatic Indexing: Classes of automatic indexing, Statistical Indexing, Natural language, Concept indexing, Hypertext linkages.

Document and Term Clustering: Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters.

UNIT-IV:

User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, weighted searches of Boolean Systems, Searching the Internet and hypertext.

UNIT-V:

Text Search Algorithms: Introduction, Software text search algorithms, Hardware text search systems.

Information System Evaluation: Introduction, Measures used in system evaluation, Measurement example – TREC results.

TEXT BOOKS:

1. Information Storage and Retrieval Systems: Theory and Implementation, Kowalski, Gerald, Mark T. Maybury, Springer
2. Modern Information Retrieval: Ricardo Baeza, Yates, Pearson Education, 2007

REFERENCES:

1. Information Retrieval: Algorithms and Heuristics, David A. Grossman and Ophir Frieder, 2nd Edition, Springer
2. Information Retrieval Data Structures and Algorithms, Frakes W. B., Ricardo Baeza-Yates, Prentice Hall, 1992

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2CS401) BIG DATA ANALYTICS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To explore and understand various HDFS commands in the Hadoop environment
- To understand the concept of the Hadoop framework and its components
- To explore various Big Data tools for data processing
- To analyze data using Visualization tools

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explore various HDFS commands on the Hadoop file system

CO-2: Illustrate Map Reduce programming framework on large data sets in a distributed environment

CO-3: Demonstrate different Big Data tools such as HIVE, PIG, and SGOOP

CO-4: Extract large data sets to perform visualization and data analysis tasks

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	1	1	1	1	1	1	1	3	3	3
CO-2	3	2	3	3	2	1	1	1	1	1	1	1	3	3	3
CO-3	3	3	2	3	3	1	1	1	1	1	1	1	3	3	3
CO-4	3	2	3	2	3	1	1	1	2	1	1	1	3	3	3

UNIT-I:

HDFS(Storage)

A. Hadoop Storage File system

Your first objective is to create a directory structure in HDFS using HDFS commands. Create the local files using Linux commands and move the files to HDFS directory and vice versa.

I. Write a command to create the directory structure in HDFS.

II. Write a Command to move file from local unix/linux machine to HDFS.

B. Viewing Data Contents, Files and Directory

Try to perform these simple steps:

a) Write HDFS command to see the contents of files in HDFS.

b) Write HDFS command to see contents of files which are present in HDFS.

C. Getting Files data from the HDFS to Local Disk

I. Write a HDFS command to copy the file from HDFS to local file system.

To process any data first move the data in HDFS. All files stored in HDFS can be accessed using HDFS commands.

Lab Instructions:

- i. Your objective is to use HDFS commands to move data to HDFS for processing data.

UNIT-II:

A. Map Reduce Programming (Processing data).

Hadoop Map-Reduce framework is developed using Java, but the framework allows you to write programs in other languages as well.

a. Word Count

The word count problem is the most famous using map reduce program. Same thing we can do with java but takes lot of time with huge file, in MR it will process less time even with huge and distributed files. The objective is to count the frequency of words of a large text.

Lab Instructions:

- i. Develop MapReduce example program in a MapReduce environment to find out the number of occurrences of each word in a text file

a. Most Frequent Words Count

Use the output from the previous program to list the most frequent words with their counts.

Lab Instructions:

- i. Use the same strategy of breaking the programs in three parts. Copy the files from the previous exercise and use them as a starting point.
- ii. Load the data from the output by using third filter to load the files.
- b) K-means clustering using map reduce

UNIT-III:

a) Data Processing Tool – Hive (NoSQL query based language)

Hive command line tool allows you to submit jobs via bash scripts.

Identifying properties of a data set:

We have a table 'user data' that contains the following fields:

data_date: string

user_id: string

properties: string

The properties field is formatted as a series of attribute=value pairs.

Ex: Age=21; state=CA; gender=M;

Lab Instructions:

- i. Create the table in HIVE using a hive NoSQL based query.
- ii. Fill the table with sample data by using some sample data bases.
- iii. Write a program that produces a list of properties with minimum value(min_value), largest value(max_value) and number of unique values. Before you start, execute the prepare step to load the data into HDFS.
- iv. Generate a count per state.

Now that extracted the properties, calculate the number of records per state.

Lab Instructions:

- i. Write a program that lists the states and their count from the data input.

b) Data Processing Tool – Pig (Latin based scripting lang)

Pig command line tool like the Hive allows you to submit jobs via bash scripts.

A. Simple Logs

We have a set of log files and need to create a job that runs every hour and perform some calculations.

The log files are delimited by a 'tab' character and have the following fields:

a) site

b) hour_of_day

- c) page_views
- d) data_date

The log files are located on the prepare folder. Load them in HDFS at data/ pig/ simple_logs folder and use them as the input.

Important: In order to load tab delimited files use pigStorage ('\u0001').

Lab Instructions:

Create a program to:

- i. Calculate the total views per hour per day.
- ii. Calculate the total views per day.
- iii. Calculate the total counts of each hour across all days.
- iv. We can write word count script by passing text file as input

UNIT-IV: SQOOP

a) It is used to import and export data from SQL to HDFS and Vice versa.

Before processing any data first I need to import data into HDFS. As of now seen importing files into HDFS but using the sqoop tool we can import SQL table data into hive table.

- I. Create table in HIVE using hive query language.
- II. Import the sql table data into hive using sqoop too.
- III. Export hive table data into local machine and into SQL.

UNIT-V:

A) Programs on Data Visualization Techniques. (R/Python)

B) Mahout machine learning library to facilitate the knowledge build up in big data analysis

TEXTBOOK:

- 1. Hadoop: The Definitive Guide: Storage and Analysis at Internet Scale by Tom White
OREILLY Publications

REFERENCES:

- 1. MapReduce Design Patterns by Donald Miner and Adam Shook OREILLY Publications.
- 2. Hadoop in 24 Hours by Sams Teach Yourself series Professional Hadoop Solutions by Boris Lublinsky, Kevin T Smith, Alexey Yakubovich
- 3. Hadoop Operations by Eric Sammer OREILLY Publications
- 4. Big Data Analytics with Hadoop 3.0 by Sridhar Alla

ONLINE RESOURCES:

- 1. <https://www.edureka.co/big-data-and-analytics>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2AM402) DEVOPS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Proficiency in at least one programming language (e.g., Python, Java, or Shell scripting), Basic understanding of REST APIs and software development concepts.

COURSE OBJECTIVES:

- To provide hands-on experience in developing backend applications, implementing CRUD operations, and exposing REST APIs
- To introduce students to version control systems using Git and GitHub, enabling effective source code management and collaboration
- To familiarize students with continuous integration and deployment (CI/CD) pipelines, using tools like CircleCI and GitHub Actions
- To equip students with containerization skills, enabling them to package and manage applications using Docker and Kubernetes
- To develop proficiency in API testing and automation, utilizing tools like Postman and writing test cases for backend services

COURSE OUTCOMES: After completion of the course, the student should be able to

CO1: Develop backend applications with CRUD operations and deploy them following DevOps best practices

CO2: Manage and collaborate on source code using Git and GitHub, demonstrating version control strategies

CO3: Set up and utilize CI/CD pipelines using CircleCI/GitHub Actions to automate builds and deployments

CO4: Containerize applications using Docker, automate deployments, and orchestrate them using Kubernetes

CO5: Perform API testing with Postman and write automated test cases to validate backend functionality

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	1	1	2	2	2	3	3	2	3
CO-2	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-3	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-4	3	3	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

1. Write code for a simple backend project. It should have basic CRUD operations and APIs (GET, POST, DELETE, PATCH/UPDATE).
2. Explore Git and GitHub commands.
3. Practice Source code management on GitHub. Experiment with the source code written in exercise 1.
4. Explore Docker commands for container management.
5. Develop a simple containerized application using Docker.
6. Integrate Kubernetes and Docker
7. Automate the process of running containerized application developed in exercise 7 using Kubernetes.
8. CircleCI/ Github Actions setup, explore the environment.
9. Demonstrate continuous integration and development using CircleCi/Github Actions
10. Install postman and explore manual API testing.
11. Develop test cases for simple APIs in the language of choice.
12. Integrate any of the suitable monitoring and logging dashboards with your application.
13. Hands-on parallel exploration of Terraform, and its configurations

TEXT BOOKS:

1. Joakim Verona. Practical Devops, Second Edition. Ingram short title; 2nd edition (2018). ISBN-10: 1788392574
2. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint. Wiley publications. ISBN: 9788126579952

REFERENCES:

1. Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley
2. Edureka DevOps Full Course - https://youtu.be/S_0q75eD8Yc

B.Tech. VIII Semester

(22PE1CS403) COMPUTER VISION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the basic concepts of computer vision
- To explore the low-level feature extraction methods
- To understand the deep learning models for computer vision
- To introduce 3D computer vision and its applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Describe the basic computer vision concepts

CO-2: Extract the feature extraction methods

CO-3: understand the deep learning models for computer vision

CO-4: Explore and identify 3D computer vision applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	1	2	1	1	1	1	1	1	1	1	1
CO-2	2	3	3	3	3	3	1	1	1	1	1	1	3	3	3
CO-3	3	3	3	3	3	3	1	1	1	1	1	1	3	3	3
CO-4	1	1	1	1	1	2	1	1	1	1	1	1	1	2	1

UNIT-I:

Digital Image Fundamentals: Elements of visual perception, light and the electromagnetic spectrum, Image sensing, and Acquisition, Sampling and Quantization, Basic relationships between pixels, Gray level transformations, Histogram processing, Spatial filtering: smoothing and sharpening.

UNIT-II:

Visual Feature Extraction: Low-level Feature Extraction: Color histograms, First order edge detection, Second order edge detection, Scale Invariant Feature Transform (SIFT), Texture features, Mathematical morphology.

UNIT-III:

Recognition: Image Recognition, Image classification, Object detection, semantic segmentation, Video understanding, vision, and language.

UNIT -IV:

Applications: Object Detection, Surveillance application, Face recognition, Activity recognition, Segmentation in autonomous driving, Robot vision

UNIT-V:

3D Computer vision: Stereo-pair images and depth perception, 3D vision systems, 3D vision applications, Case examples: 3D systems for vision impaired persons, face and body modeling, clinical and veterinary applications, movie restoration.

TEXTBOOKS:

1. Digital Image Processing, Gonzalez R. C., Pearson Education, 2009
2. Feature Extraction and Image Processing for Computer Vision, Nixon M. & Aguado A., Academic Press, 2019
3. Deep Learning in Computer Vision: Principles and Applications, Hassaballah M., & Awad A. I. (Eds.), CRC Press, 2020

REFERENCES:

1. An Introduction to 3D Computer Vision Techniques And Algorithms, Cyganek B., & Siebert J. P., John Wiley & Sons, 2011
2. Computer Vision: Algorithms and Applications, Szeliski R., Springer Nature, 2022

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1AM405) COGNITIVE ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To identify the basics of cognitive science engineering with focus on mind organization, brains
- To analyze human intelligence and artificial intelligence
- To understand basics of neuroscience and linguistics
- To analyze various levels of cognitive functioning and its applications
- To relate the concepts of cognitive field in robotics and real time applications

COURSE OUTCOMES: After Completion of the course the student is able to

CO-1: Introduce role of computers in cognitive science and neuroscience applications

CO-2: Illustrate various information processing models and cognitive architecture

CO-3: Apply various computation tools for cognitive functioning

CO-4: Analyze cognitive science applications in image processing linguistic applications

CO-5: Demonstrate the use of cognitive models and patterns in distributed environment, and real time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	2	1	1	1	2	1	1	1	2	2	2
CO-2	2	3	2	2	3	1	1	2	2	2	1	1	3	3	2
CO-3	2	3	3	2	2	1	2	1	2	2	2	1	3	3	3
CO-4	3	3	3	2	2	2	1	2	3	3	2	2	3	3	2
CO-5	2	2	2	3	2	1	2	1	2	2	3	2	2	2	3

UNIT-I:

Introduction to Cognitive Science: The Cognitive view –Some Fundamental Concepts – Computers in Cognitive Science – Applied Cognitive Science – The Interdisciplinary Nature of Cognitive Science, The Architecture of the Mind. Cognitive Processes, Computational Neuroscience - The Organization of the mind.

UNIT-II:

Higher-Level Cognition and Challenges: Dynamical systems and situated cognition-Challenges – Emotions and Consciousness – Computation of Cognitive Functioning in machines: Robotics, Human-Robotics Interaction.

UNIT-III:

Cognitive Science for Vision/Image Processing: Perception and sensing: visual cognition, cognitive mechanisms of vision, Feature Extraction from Images,

Information Processing in Perception and Visual Behavior, Cognitive function measurement tools and software.

UNIT-IV:

Language Acquisition and Semantics: Milestones in acquisition, Theoretical perspectives, Semantics and Cognitive Science, Meaning and entailment, Sense, Problems in possible worlds Semantics, Cognitive and computational models of semantic Processing.

UNIT-V:

Distributed Cognitive Learning Using Fuzzy Cognitive Maps: Axelrod's cognitive maps, Kosko's model, adaptive FCM, Pal and Konar's FCM model-examples.

TEXT BOOKS:

1. Cognitive Science: An Introduction, Neil Stillings, Steven E. Weisler, Christopher H. Chase and Mark H. Feinstein, 2nd Edition, 1995
2. Cognitive Science: An Introduction to the Science of the Mind, José Luis Bermúdez, Cambridge University Press, 2010
3. Cognitive Engineering: A Distributed Approach to Machine Intelligence (Advanced Information and Knowledge Processing), Amit Konar, Springer, 2005

REFERENCES:

1. Computational Vision: Information Processing in Perception and Visual Behavior, Hanspeter A. Mallot, Translated by John S. Allen
2. How the Mind Works, Steven Pinker, 2009
3. Cognitive Science: An Interdisciplinary Approach, Carolyn Panzer Sobel and Paul Li, 2013
4. Mind: Introduction to Cognitive Science, Paul Thagard, 2nd Edition, MIT Press, 2004
Wilson Robert A., & Keil Frank C (eds.)
5. The MIT Encyclopedia of the Cognitive Sciences (MITECS), MIT Press, 2001

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CS404) AUGMENTED REALITY AND VIRTUAL REALITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Programming and Data Structures

COURSE OBJECTIVES:

- To understand the basic concepts of Virtual and Augmented Reality
- To identify the differences in AR/VR concepts and technologies
- To understand the fundamental concepts relating to Virtual Reality such as presence, immersion, and engagement
- To evaluate usability of AR/VR applications and critique their use of AR/VR capabilities
- To design and prototype effective AR/VR applications using UNITY platform for business, industry, non-profit and government organizations

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Introduce role of augmented and virtual reality in interactive application development

CO-2: Apply Geometric Modeling Techniques

CO-3: Implement virtual environment with respect to user experience

CO-4: Design Virtual/Augmented Reality Applications

CO-5: Acquire fundamental and hands-on skills needed to develop AR/VR applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	1	1	2	2	1	-	2	2	3	3	3
CO-2	1	1	1	3	1	2	2	1	2	1	1	2	3	3	3
CO-3	1	1	2	-	3	1	1	1	1	2	1	2	3	3	3
CO-4	-	2	-	-	-	2	-	2	1	1	3	1	3	3	3
CO-5	-	-	2	-	3	-	-	-	-	-	-	-	3	3	3

UNIT-I:

Introduction to Augmented Reality (AR): Definition and Scope A Brief History of Augmented Reality Displays (Multimodal Displays, Spatial Display Model, and Visual Displays) Strong vs Weak AR Applications of AR Challenges in AR

UNIT-II:

Introduction to Virtual Reality (VR): Definition and Scope Types of VR Characteristics of VR Basic VR environments Limitations of VR environments Immersion Vs Presence Key hardware requirements for VR

UNIT-III:

Interaction Design for AR/VR Environments: Interaction design process Identifying user needs AR/VR design considerations Typical AR/VR Interface Metaphors Affordances in AR/VR Human Information Processing Design for Perception and cognition User experience (UX) guidelines for AR/VR UX challenges for AR/VR Prototyping for AR/VR Evaluation of the developed AR/VR prototype

UNIT-IV:

Unity Overview: Windows, Interface, Navigation, Terminology, Game Objects, Hierarchy, Parenting Objects Asset Store, Importing Plug-ins Creating a Terrain, Materials, Colors, Transparency Introduction to MonoBehaviour: Awake, Start, Update

UNIT-V:

Introduction to Vuforia and Physics in UNITY

Vuforia Overview: Interface, Navigation, Terminology, Image Targeting, Custom Images Overview of Physics in Unity

Introduction to Scripting: Terminology, Creating Objects, Accessing Components, Debugging, Lists, Loops

TEXT BOOKS:

1. Understanding virtual reality, Sherman W. R., & Craig A. B., Morgan Kauffman, 2003
2. Augmented Reality: Principles and Practice, Schmalstieg D., & Hollerer T., Addison-Wesley Professional, 2016

REFERENCES:

1. A survey of augmented reality. Presence: Teleoperators & Virtual Environments, Azuma R. T., 1997, 6(4), 355–385
2. Recent Advances in Augmented Reality, IEEE Computer Graphics and Applications, Azuma R., Baillot Y., Behringer R., Feiner S., Julier S., & MacIntyre B., 2001, 21(6), 34-47
3. A Cost-Effective Interactive 3D Virtual Reality System Applied To Military Live Firing Training, Virtual Reality, Bhagat K. K., Liou, W. K., & Chang, C. Y, 2016, 20(2), 127-140. doi:10.1007/s10055-016-0284-x
4. Augmented Reality Technologies, Systems And Applications, Multimedia Tools and Applications, Carmigniani J., Furht B., Anisetti M., Ceravolo P., Damiani E., & Ivkovic M., 2011, 51(1), 341-377
5. Human Augmentation: Past, Present and Future. International Journal Of Human Computer Studies, Raisamo R., Rakkolainen I., Majaranta P., Salminen K., Rantala J., & Farooq, 2019, 131, 131-143. doi: <https://doi.org/10.1016/j.ijhcs.2019.05.008>

ONLINE RESOURCES:

1. Virtual Reality Engineering, Manivannan M., 2018, IIT Madras, <https://nptel.ac.in/courses/121106013>
2. Industry 4.0: Augmented Reality and Virtual Reality, Misra S., 2019, IIT Kharagpur, <https://www.youtube.com/watch?v=zLMgdYI82IE>
3. Augmented Reality - Fundamentals and Development, NPTEL Special Lecture Series, Dube A., 2020, <https://www.youtube.com/watch?v=MGuSTAqIZ9Q>
4. <http://cambum.net/course-2.htm>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PC1CY401) BLOCKCHAIN TECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To provide an understanding of blockchain technology, including its origin, components, types, consensus mechanisms, and cryptocurrency applications
- To explore public blockchain systems and smart contracts, their principles, implementations, and industry applications
- To know the security issues of blockchain technology
- To understand security challenges in blockchain and explore industry applications.
- To study practical blockchain implementations through case studies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze blockchain technology by examining its origin, components, types, consensus mechanisms, and cryptocurrency applications

CO-2: Evaluate public blockchains and smart contracts, their principles, implementations, and industry applications

CO-3: Examine private and consortium blockchains, their applications, and the role of ICOs in blockchain fundraising

CO-4: Apply blockchain security principles and explore its applications, limitations, and future scope

CO-5: Implement blockchain solutions through case studies and development with Python and Hyperledger Fabric

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	1	1	1	2	1	1	1	2	2	2	2
CO-2	3	2	3	3	3	2	2	3	2	1	3	3	3	3	2
CO-3	2	2	2	1	1	1	1	2	1	1	1	2	3	3	3
CO-4	3	3	2	1	1	1	1	2	1	1	2	3	3	3	2
CO-5	3	3	3	1	1	1	1	1	1	1	1	2	2	2	3

UNIT-I:

Fundamentals of Blockchain: Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency – Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.

UNIT-II:

Public Blockchain System: Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts:

Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.

UNIT-III:

Private Blockchain System: Introduction, Key Characteristics of Private Blockchain, Why We Need Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain.

Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Why We Need Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda. Initial Coin Offering: Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering, Successful Initial Coin Offerings, Evolution of ICO, ICO Platforms.

UNIT-IV:

Security in Blockchain: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric.

Applications of Blockchain: Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.

UNIT-V:

Blockchain Case Studies: Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities.

Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.

TEXT BOOK:

1. Block Chain Technology, Chandramouli Subramanian, Asha A. George, Abhilash K. A. and Meena Karthikeyan, Universities Press

REFERENCES:

1. Blockchain Blueprint for Economy, Melanie Swan, SPD O'Reilly
2. Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gaur, Pearson Addison Wesley

ONLINE RESOURCES:

1. The Bitcoin Backbone Protocol - Analysis and Applications, J. A. Garay et al., EUROCRYPT2015LNCSVOI9057, Vol. II, pp. 281310 (eprint.iacr.org/2016/1048)

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1IT403) ADVERSARIAL MACHINE LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PREREQUISITES: Basic knowledge of Machine Learning

COURSE OBJECTIVES:

- To understand the basic notation and introduce the framework for secure learning and the various phases involved in the learning process
- To study causative attacks on machine learning algorithms and their impact on learning
- To examine a case study on causative attacks against the SpamBayes spam filter and understand the defense mechanisms used
- To explore different types of attacks on specific machine learning models such as hypersphere learners and PCA detectors
- To delve into exploratory attacks on machine learning, focusing on privacy-preserving mechanisms for SVM learning

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basic notation and how security analysis fits into the learning process

CO-2: Apply the gained knowledge to defend machine learning models against causative attacks

CO-3: Create a practical experience in corrupting PCA subspaces

CO-4: Evaluate the SpamBayes spam filter and its susceptibility to causative attacks

CO-5: Analyse privacy-preserving mechanisms for SVM learning in protecting sensitive data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	2	1	1	2	1	2	-	-	-	1	2	1	2
CO-2	2	2	3	2	2	3	1	3	-	-	-	2	3	2	3
CO-3	2	2	2	3	3	1	2	2	-	-	-	2	3	3	2
CO-4	2	2	3	2	2	2	1	3	-	-	-	2	2	3	2
CO-5	2	2	3	2	2	2	1	1	-	-	-	2	3	2	3

UNIT-I:

Introduction

Background and Notation: Basic Notation, Statistical Machine Learning.

A Framework for Secure Learning: Analysing the Phases of Learning, Security Analysis, Framework, Exploratory Attacks, Causative Attacks.

UNIT-II:**Causative Attacks on Machine Learning**

Attacking a Hypersphere Learner: Causative Attacks on Hypersphere Detectors, Hypersphere Attack Description, Optimal Unconstrained Attacks, Imposing Time Constraints on the Attack, Constrained Attacks.

UNIT-III:

Availability Attack Case Study: Spam Bayes: The SpamBayes Spam Filter, Threat Model for SpamBayes, Causative Attacks against SpamBayes' Learner, the Reject on Negative Impact (RONI) Defense, Experiments with SpamBayes.

UNIT-IV:

Integrity Attack Case Study: PCA Detector: PCA Method for Detecting Traffic Anomalies, Corrupting the PCA Subspace, Corruption-Resilient Detectors, Empirical Evaluation.

UNIT-V:**Exploratory Attacks on Machine Learning**

Privacy-Preserving Mechanisms for SVM Learning: Privacy Breach Case Studies, Problem Setting: Privacy-Preserving Learning, Support Vector Machines: A Brief Primer.

TEXT BOOK:

1. Adversarial Machine Learning, Anthony D. Joseph, Blaine Nelson, Benjamin I. P. Rubinstein, J. D. Tygar, Cambridge University Press, 2019

REFERENCES:

1. Adversarial Machine Learning, Vol. 12, Vorobeychik, Yevgeniy, Murat Kantarcioglu, Ronald Brachman, Peter Stone, and Francesca Rossi, San Rafael, Morgan & Claypool Publishers, 2018
2. Generative Deep Learning: Teaching Machines to Paint, Foster David, Write, Compose, and Play (Japanese Version) O'Reilly Media, 2019, 139-140

ONLINE RESOURCES:

1. <https://www.udacity.com/blog/2023/03/the-role-of-ai-and-ml-in-cybersecurity.html>
2. <https://www.youtube.com/watch?v=3-qazNQS2JU>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1DS405) TIME SERIES ANALYSIS AND FORECASTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn basic analysis of time series data and time series regression
- To learn auto-regressive and model averaging models
- To learn basic concepts of forecasts for a time series data
- To learn how to perform the advanced computation like Multivariate Time Series Models and Forecasting

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply ideas to real time series data and interpret outcomes of analyses

CO-2: Learn how to construct a time-series plot and identifying the underlying patterns in the data

CO-3: Learn basic concepts factorization techniques in time series analysis

CO-4: Learn how to develop forecasts for a time series that has a seasonal pattern

CO-5: Demonstrate an advanced understanding of the underlying concepts in the time series and frequency domains for computation, visualization, and analysis of time series data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	3	3	2	-	-	2	2	2	3	3	2	1
CO-2	3	2	3	2	3	3	-	-	2	2	2	3	2	1	2
CO-3	3	2	3	3	3	3	-	-	3	2	3	2	3	2	2
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	3	2	3
CO-5	3	2	2	3	2	3	-	-	2	2	3	2	3	2	3

UNIT-I:

Introduction of time series analysis, The Nature of Time Series Data, Internal structures of time series, Time Series Statistical Models, Moving Window for Time series data.

UNIT-II:

Fourier decomposition, deep learning features LSTM, Image Histogram, Deep learning feature CNN, Relational data, Graph data, Indicator Variable, Feature binning, interaction variable.

UNIT-III:

Mathematical transforms, Model specific featurizations, Feature orthogonality, Domain specific featurizations, Feature slicing.

Kaggle Winner Solutions: Students are suggested to carry out with at least 3 Kaggle winner competition case study solutions and present the report for that.

(<https://www.kaggle.com/code/sudalairajkumar/winning-solutions-of-kaggle-competitions>)

UNIT-IV:

Autocorrelation and Partial autocorrelation, General Approach to Time Series Modelling and Forecasting, Evaluating and Monitoring Forecasting Model Performance, Introduction Least Squares Estimation in Linear Regression Models, Statistical Inference in Linear Regression.

UNIT-V:

Model Adequacy Checking, Regression Models for General Time Series Data. Exponential Smoothing, First order and Second order, Multivariate Time Series Models and Forecasting, Multivariate Stationary Process, ARIMA model, Vector AR (VAR) Models.

TEXT BOOKS:

1. Introduction to Time Series Analysis and Forecasting, Douglas C. Montgomery, Cheryl L. Jen, 2nd Edition, Wiley Series in Probability and Statistics, 2015
2. Master Time Series Data Processing, Visualization, and Modeling Using Python, Dr. Avishek Pal, Dr. P. K. S. Prakash, 2017

REFERENCES:

1. Introduction to Time Series Analysis, P. J. Brockwell and R. A. Davis
2. Time Series Analysis and Its Applications, Robert H. Shumway and David S. Stoffer
3. Introduction to Statistical Time Series, W. A. Fuller
4. Time Series Analysis, Wilfredo Palma

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1DS406) QUANTUM COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basics of quantum computation and quantum mechanics
- To apply different models of quantum computation
- To illustrate the quantum models using machine learning algorithms
- To explore various Quantum algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explore the quantum computation environment

CO-2: Identify the implications of quantum computation in the field of mathematics and associated fields

CO-3: Demonstrate building blocks and programming methodologies for quantum computing

CO-4: Apply quantum computation algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	3	2	-	-	2	2	2	3	1	-	1
CO-2	2	2	3	2	3	3	-	-	2	2	2	3	2	-	1
CO-3	2	2	3	3	3	3	-	-	3	2	3	2	1	1	1
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	1	-	1

UNIT-I:

Introduction and Background: Overview, The Circuit Model of Computation, A Linear Algebra Formulation of the Circuit Model, Reversible Computation. A Preview of Quantum Physics.

Linear Algebra and The Dirac Notation: The Dirac Notation and Hilbert, Dual Vectors, Operators, The Spectral Theorem, Functions of Operators

UNIT-II:

Qubits and The Framework of Quantum Mechanics: The State of a Quantum System, Time-Evolution of a Closed System, Composite Systems.

A Quantum Model of Computation: The Quantum Circuit Model, Quantum Gates, 1-Qubit Gates, Controlled-U Gates, Universal Sets of Quantum Gates, Efficiency of Approximating Unitary Transformations.

UNIT-III:

Superdense Coding and Quantum Teleportation: Superdense Coding, Quantum Teleportation, An Application of Quantum Teleportation.

Quantum Error Correction: Error Models for Quantum Computing, Encoding, Error Recovery. Architecture: Bits and Qubits, Classical Gates, Reversible Gates, Quantum Gates.

UNIT-IV:

Programming Languages: Programming in a Quantum World, Quantum Assembly Programming, Toward Higher-Level Quantum Programming.

Theoretical Computer Science: Deterministic and Nondeterministic Computations, Probabilistic Computations, Quantum Computations.

UNIT-V:

Quantum Algorithms: Deutsch's Algorithm, The Deutsch–Jozsa Algorithm, Simon's Periodicity Algorithm, Grover's Search Algorithm, Shor's Factoring Algorithm

TEXTBOOKS:

1. An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Michele Mosca, Oxford University Press
2. Quantum Computing for Computer Scientists, Noson S. Yanofsky, Michael A. Mannucci, Cambridge University Press

REFERENCES:

1. Quantum Computing, Ronald De Wolf, QuSoft, CWI and University of Amsterdam
2. Quantum Computing Explained, David McMahon, Wiley
3. Quantum Computing Fundamentals, William (Chuck) Easttom

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CY410) ETHICAL HACKING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Operating Systems, Computer Networks, Network Security & Cryptography

COURSE OBJECTIVES:

- To understand the basic concepts of ethical Hacking
- To learn the enumeration and system Hacking process
- To understand the threats of Trojans, backdoors, and sniffers
- To learn session hijacking and its various techniques, as well as DoS, DDoS, and botnet attacks
- To learn to defend against database attacks and understand social engineering techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand ethical hacking and footprinting process

CO-2: Analyze various enumeration and system hacking methods

CO-3: Identify, analyze, and counter cybersecurity risks posed by Trojans, backdoors, and sniffers

CO-4: Apply various techniques and methods used in session hijacking, Dos, DDOS and Botnets

CO-5: Detect and mitigate SQL injection vulnerabilities and implement effective policies to counter social engineering risks

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	3	1	3	1	1	3	3	3	1	2	3	3	3
CO-2	2	1	3	2	3	3	3	3	2	1	2	2	3	3	3
CO-3	2	2	3	3	3	3	2	3	1	1	2	2	3	3	3
CO-4	1	2	3	3	3	3	2	3	1	1	2	2	3	3	3
CO-5	1	1	3	2	2	3	3	3	3	3	2	2	3	3	3

UNIT-I:

Ethical Hacking Basics: Security Fundamentals, Security Testing, Hacker and Cracker Descriptions, Ethical Hackers, Test Plans—Keeping It Legal, Ethics and Legality, The Attacker's Process, The Ethical Hacker's Process,

Footprinting and Scanning: The Seven-Step Information-Gathering Process, Identifying Active Machines, Finding Open Ports and Access Points, OS Fingerprinting.

UNIT-II:

Enumeration: Windows Enumeration, Windows Security, NetBIOS and LDAP Enumeration, SNMP Enumeration, Linux/UNIX Enumeration, NTP Enumeration, SMTP Enumeration,

System Hacking: Nontechnical Password Attacks, Technical Password Attacks, Password Guessing, Automated Password Guessing. Password Sniffing, Keystroke Loggers, Privilege Escalation and Exploiting Vulnerabilities, Exploiting an Application, Buffer Overflow, Owning the Box, Cracking the Passwords, Hiding Files and Covering Tracks, File Hiding.

UNIT-III:

Trojans and Backdoors: Trojan Types, Trojan Ports and Communication Methods, Trojan Goals, Trojan Infection Mechanisms, Effects of Trojans, Trojan Tools, Distributing Trojans, Trojan Tool Kits, Covert Communication and Tools, Port Redirection, Other Redirection and Covert Tools, Keystroke Logging and Spyware, Hardware, Software, Spyware, Trojan and Backdoor Countermeasures

Sniffers: Passive Sniffing, Active Sniffing, Address Resolution Protocol, ARP Poisoning and Flooding, Tools for Sniffing, Wireshark, Other Sniffing Tools, Sniffing and Spoofing Countermeasures

UNIT-IV:

Session Hijacking: Transport Layer Hijacking, Predict the Sequence Number, Take One of the Parties Offline, Take Control of the Session, Application Layer Hijacking, Session Sniffing, Predictable Session Token ID, Man-in-the-Middle Attacks, Man-in-the-Browser Attacks, Client-Side Attacks, Session-Hijacking Tools, Preventing Session Hijacking Denial of Service, Distributed Denial of Service, and Botnets: Types of DoS, Bandwidth Attacks, SYN Flood Attacks, Program and Application Attacks, Distributed Denial of Service, DDoS Tools, Botnets, Countermeasures of DoS, DDOS, and Botnet

UNIT-V:

Database Attacks Database Hacking, Identifying SQL Servers, SQL Injection Vulnerabilities, SQL Injection Hacking Tools.

Social Engineering: Types of Social Engineering, Person-to-Person Social Engineering, Computer-Based Social Engineering, Reverse Social Engineering, Policies and Procedures, Employee Hiring and Termination Policies, Help Desk Procedures and Password Change Policies, User Awareness.

TEXT BOOKS:

1. Certified Ethical Hacker (CEH) Cert Guide, Michael Gregg, 3rd Edition, Pearson, 2019
2. Hands-On Ethical Hacking and Network Defense, Michael T. Simpson, Kent Backman, James E. Corley, 3rd Edition, Cengage Learning, 2017

REFERENCES:

1. Ethical Hacking and Countermeasures Attack Phases, EC-Council, Cengage Learning
2. CEH: Official Certified Ethical Hacker Review Guide, Kimberly Graves, Wiley, 2007

B.Tech. VIII Semester

(22PE1AM407) ROBOTIC PROCESS AUTOMATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explore the knowledge of basic concepts of Robotic Process Automation
- To outline the RPA design & development strategies
- To apply design methodologies adopted by UiPath
- To develop flow diagrams using RPA for real-world application
- To apply programming robots for logging and exception handling

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand Robotic Process Automation concepts

CO-2: Adopt UiPath programming techniques to deploy robot configurations

CO-3: Explore various data extraction techniques and perform integrations with various popular applications

CO-4: Design and develop a programmed robot that includes logging and exception handling

CO-5: Deploy and control Bots with UiPath orchestrator

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	3	3	2	2	2	1	1	2	2	2	2	1
CO-2	2	2	2	1	3	1	1	2	1	1	1	1	3	2	2
CO-3	1	2	2	3	2	2	1	3	1	2	1	2	3	2	3
CO-4	2	2	3	3	2	2	1	2	1	2	1	2	2	2	3
CO-5	1	2	1	1	3	1	1	2	1	2	1	1	2	3	3

UNIT-I:

Introduction: What is Robotic Process Automation (RPA), Scope & techniques of Automation, Benefits of RPA, Components of RPA, RPA Platforms, UiPath Studio, Installation of UiPath Studio, Learning UiPath Studio.

UNIT-II:

Sequence, Flowchart & Control Flow: Sequencing the Workflow, Activities, Flowchart, Control Flow for Decision making. Data Manipulation: Variables, Collection, Arguments, Data Table, Clipboard management, File operations with step-by-step example

Controls: Finding the control, waiting for a control, Act on a control, UiExplorer, Handling Events Recording and advanced UI Interaction Basic recording, Desktop recording, Web recording, Citrix, Screen Scraping, when to use OCR, Types of OCR available, how to use OCR Avoiding typical failure points.

UNIT-III:

Plugins and Extensions: Terminal plugin, Java plugin, Java plugin with UiPath Studio, Citrix automation, Citrix environment, Mail plugin, PDF plugin, Web integration, Excel and Word plugins, Credential management Extensions Handling User Events and Assistant Bots: What are assistant bots, Monitoring system event triggers: Hotkey trigger, Mouse trigger, System trigger, Monitoring image and element triggers, Launching an assistant bot on a keyboard event.

UNIT-IV:

Exception Handling, Debugging, and Logging: Exception handling, Common exceptions and ways to handle them, Logging and taking screenshots Debugging techniques, setting break points, Slow step, Highlighting, Break, collecting crash dumps: Enabling crash dumps, Disabling crash dumps, Error reporting: Enterprise Edition customers, Community Edition users. Managing and Maintaining the Code: Project organization, picking an appropriate layout for each workflow, Breaking the process into smaller parts, Using exception handling, Making your workflow readable, Keeping it clean, Nesting workflows, Reusability of workflows, Templates, Commenting techniques, State Machine, When to use Flowcharts State Machines or Sequences.

UNIT-V:

Deploying and Maintaining the Bot: Publishing using publish utility, Overview of Orchestration Server, Using Orchestration Server to control bots, Using Orchestration Server to deploy bots, License management, Activating and uploading a license to Orchestrator, Publishing and managing updates, Packages, Managing packages.

TEXT BOOK:

1. Learning Robotic Process Automation, Alok Mani Tripathi, O'Reilly Publishing, 2018

REFERENCES:

1. Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant, Richard Murdoch, Amazon Asia-Pacific Holdings, 2018
2. Introduction to Robotic Process Automation: A Primer, Institute of Robotic Process Automation, Frank Casale, Rebecca Dilla, Heidi Jaynes, Lauren Livingston, Amazon Asia-Pacific Holdings, 2018

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CS405) SOCIAL MEDIA ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Graph Theory, Data Mining, Python/R programming

COURSE OBJECTIVES:

- To familiarize with the concept of social media
- To familiarize with the concept of social media analytics and understand its significance
- To enable to develop skills required for analyzing the effectiveness of social media
- To examine the ethical and legal implications of leveraging social media data
- To familiarize with different visualization techniques for Social media analytics

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the concept of social media Analytics and its significance

CO-2: Analyze the effectiveness of social media

CO-3: Use different effective Visualization techniques to represent social media analytics

CO-4: Use different Social media analytics tools effectively and efficiently

CO-5: Acquire the fundamental perspectives and hands-on skills needed to work with social media data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	3	3	2	-	-	2	2	2	3	3	3	3
CO-2	3	2	3	2	3	3	-	-	2	2	2	3	3	3	3
CO-3	3	2	3	3	3	3	-	-	3	2	3	2	3	3	3
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	3	2	2
CO-5	3	2	2	3	2	3	-	-	2	2	3	2	3	2	3

UNIT-I:

Social Media Analytics: An Overview: Core Characteristics of Social Media, Types of Social Media, Social media landscape, Need for Social Media Analytics (SMA), SMA in small & large organizations. Purpose of Social Media Analytics, Social Media vs. Traditional Business Analytics, Seven Layers of Social Media Analytics, Types of Social Media Analytics, Social Media Analytics Cycle, Challenges to Social Media Analytics, Social Media Analytics Tools

UNIT-II:

Social Network Structure, Measures & Visualization: Basics of Social Network Structure - Nodes, Edges & Tie Describing the Networks Measures - Degree Distribution, Density, Connectivity, Centralization, Tie Strength & Trust Network Visualization - Graph Layout, Visualizing Network features, Scale Issues. Social Media Network Analytics - Common

Network Terms, Common Social Media Network Types, Types of Networks, Common Network Terminologies, Network Analytics Tools.

UNIT-III:

Social Media Text, Action & Hyperlink Analytics: Social Media Text Analytics - Types of Social Media Text, Purpose of Text Analytics, Steps in Text Analytics, Social Media Text 8 Analysis Tools Social Media Action Analytics - What Is Actions Analytics? Common Social Media Actions, Actions Analytics Tools Social Media Hyperlink Analytics - Types of Hyperlinks, Types of Hyperlink Analytics, Hyperlink Analytics Tools

UNIT-IV:

Social Media Location & Search Engine Analytics: Location Analytics - Sources of Location Data, Categories of Location Analytics, Location Analytics and Privacy Concerns, Location Analytics Tools Search Engine Analytics - Types of Search Engines, Search Engine Analytics, Search Engine Analytics Tools

UNIT-V:

Social Information Filtering: Social Information Filtering - Social Sharing and filtering, Automated Recommendation systems, Traditional Vs social Recommendation Systems Understanding SocialMedia and Business Alignment, Social Media KPI, Formulating a Social Media Strategy, Managing Social Media Risks

TEXT BOOKS:

1. Seven Layers of Social Media Analytics - Mining Business Insights from Social Media Text, Actions, Networks, Hyperlinks, Apps, Search Engine, and Location Data, Gohar F. Khan, 2015
2. Mining the Social Web - Analyzing Data from Facebook, Twitter, LinkedIn, and Other Social Media Sites, Matthew A. Russell, O'Reilly
3. Social Network Data Analytics, Charu Aggarwal (Ed.), Springer, 2011

REFERENCES:

1. Analyzing the Social Web, Jennifer Golbeck, 1st Edition
2. Social Media Analytics, Techniques and Insights for Extracting Business Value Out of Social Media, Matthew Ganis, Avinash Kohirkar, IBM Press, 2015
3. Social Media Analytics Strategy - Using Data to Optimize Business Performance, Alex Gonçalves, APress
4. Social Media Data Mining and Analytics, Szabo G., G. Polatkan, O. Boykin & A. Chalkiopoulos, Wiley, 2019
5. Social Media Analytics: Effective Tools for Building, Interpreting and Using Metrics, Marshall Sponder, 1st Edition, McGraw-Hill, 2011

ONLINE RESOURCES:

1. <https://cse.iitkgp.ac.in/~pawang/courses/SC16.html>
2. https://onlinecourses.nptel.ac.in/noc20_cs78/preview